MILITARY SPECIFICATION

CAPSULE AND PESTLE, DENTAL AMALGAMATOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a capsule with pestle suitable for amalgamating silver alloy pellets in a mechanical amalgamator (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, Attn: Directorate of Medical Materiel, CODE ATT, 2800 S. 20th St., Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Material. The capsule shall be injection molded from a suitable grade of gray-colored, surface-lubricated acetal resin (see 6.2). The pestle shall be machined from suitable gray-colored teflon rod.

3.2 Figure. The figure forming a part of this specification is not a production drawing. This figure establishes dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitation- prescribed on the figure to provide a product with proper design, configuration and function. No deviation from the dimensional limitations prescribed on the figure is permissible without prior approval of the contracting officer.

3.3 Style and design. The product shall consist of a capsule with pestle. The capsule and pestle shall be suitable for amalgamating silver alloy pellets and shall be suitable for attachment to, and use with, dental electric amalgamators (see 6.3). The capsule and pestle shall be designed so as to permit pre-loading of dental mercury and silver alloy pellets. Style and design of capsule and pestle shall be as shown in figure 1.

3.4 Construction and dimensions.

3.4.1 Capsule. The capsule shall be of two-piece construction and shall consist of a cap and a body. When the cap and the body are assembled, the resulting capsule cavity shall be 1 $\pm$ 1/16 inch long by 0.390 $\pm$ 0.010 inch in diameter, and the inside ends shall be molded to a radius of 5/8 $\pm$ 1/64 inch. The outside diameter of the capsule shall be 1/2 to 17/32 inch, and the overall assembled length shall be 1-1/4 $\pm$ 1/16 inches. The outside diameter of the body shall be reduced for a 3/16-inch length at the open end to provide an external shoulder, and the inside diameter of the cap shall be enlarged to provide an internal shoulder for proper mating of cap and body. When the capsule is completely assembled, the lip of the cap shall clear the external shoulder of the body by 0.008 $\pm$ 0.002 inch. The capsule shall fit in a capsule holder (see 6.4).

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3.4.2 Pestle. The pestle shall be a barbell design. The pestle shall be 0.535 ± 0.025 inch in length, 0.300 ± 0.010 inch in diameter at the enlarged ends, and 0.150 ± 0.008 inch in diameter at the center section. The weight of the pestle shall be 0.63 ± 0.02 grams.

3.5 Finish. The capsules and pestles shall have a smooth finish, except for the knurled or ridged gripping surface on the capsule. The cap and the body shall be knurled or ridged for a minimum of 3/16 inch to prevent slippage of the capsule in the fingers. A knockout or indentation at the ends or top of the capsule body will be acceptable, if required during the manufacturing process.

3.6 Workmanship. Workmanship shall be first class throughout. The capsule and pestle shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of compliance. Certificates of compliance, supplied by the manufacturer of the acetal resin and teflon, may be furnished in lieu of actual performance of such testing by the contractor. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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FIGURE 1 CAPSULE AND PESTLE

Dimensions shall be as specified in 3.4.1 and 3.4.2.

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4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in tables I, II and III (see 4.2.1, 4.2.2 and 4.3) and 4.3.1.

4.2.1 Sampling for examination. A random sample of capsules and pestles shall be selected from each lot in accordance with MIL-STD-105. The inspection level and Acceptable Quality Level (AQL) shall be as indicated in table I. Unit of product for examination purposes shall be one capsule and pestle.

TABLE I. Examination criteria.

Type defect	Inspection level	AQL (percent defective)
Major	II	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II, except that the acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be one capsule with pestle.

TABLE II. Test criteria.

Characteristic	Requirement	Test procedure	Inspection level
Weight of pestle	3.4.2	4.4.1	S-2

4.3 Examination. The capsule and pestle shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table III.

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TABLE III. Classification of defects.

Categories	Defects
Major:	
101	Style and design of capsule and pestle not as specified.
102	Construction of capsule and pestle not as specified.
103	Inside dimensions of capsule, and dimensions of pestle, not as specified.
104	Finish of capsule and pestle not as specified.
Minor A:	
201	Color of capsule and pestle not as specified.
Minor B:	
251	Dimensions (other than 103) not as specified.

4.3.1 Packaging inspection. The inspection of the packaging, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and marking requirements of MIL-STD-129.

4.4 Test procedures.

4.4.1 Weight of pestle. The weight of the pestle shall be determined by a scale suitable for the purpose intended.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package, containing twelve (12) capsules and pestles as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit packaging. One unit of issue shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

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5.1.1.1.2 Intermediate packaging. Twelve (12) units of issue shall be packaged in a box of appropriate size and design, constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.1.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. One hundred and forty four (144) units of issue shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with the appendix of PPP-B-636.

5.2.2 Level B. One hundred and forty four (144) units of issue shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior and exterior containers. Each unit, intermediate and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 Suitable acetal resins are "Celcon M90-02", as manufactured by Celanese Plastics Company, a division of Celanese Corporation and "Delrin 500 and 900", as manufactured by E. I. Dupont de Nemours and Company, Inc.

6.3 The capsule with pestle covered by this specification is intended for use with dental electric amalgamators NSN's 6520-00-500-5010 and 6520-00-139-4580.

6.4 The capsule covered by this specification is intended for use with capsule holder NSN 6520-00-889-6565.

6.5 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

NSN	Item Identification
6520-00-889-5727	CAPSULE AND PESTLE, DENTAL AMALGAMATOR, 12s

Military Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1413

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-37360 CAPSULE AND PESTLE, DENTAL AMALGAMATOR

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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